

stockplotr: :CHEATSHEET

Create report-ready tables and figures for stock assessments.



The Basics

Convert output file to a standardized framework.

```
data <- convert_output(  
  file = <FILE_PATH> ,  
  model = "MODEL ACRONYM" ,  
  fleet_names = c("FLEET_NAMES") ,  
  save_dir = <FILE_PATH>  
)
```

Optional

label	estimate	year	fleet	sex	...
spawning_biomass	2017090	2016	NA	NA	...
spawning_biomass	2286740	2017	NA	NA	...
...	...	...	...	...	...

Create customizable plots.

```
plot_biomass(  
  dat = <DATA> ,  
  geom = <GEOM> ,  
  group = "<COLUMN_NAME>" ,  
  facet = "<COLUMN_NAME>" ,  
  ref_line = "msy" ,  
  unit_label = "metric tons" ,  
  module = "MODULE_NAME" ,  
  scale_amount = 1 ,  
  relative = FALSE ,  
  make_rda = FALSE ,  
  figures_dir = <FILE_PATH> ,  
  interactive = TRUE ,  
  ...  
)
```

"line", "point" or "area"

Export object as an .rda

Additional ggplot2 arguments

Example

```
plot_biomass(example_data, unit_label = "mt", ref_line = "target",  
  scale_amount = 100, module = "TIME_SERIES",  
  make_rda = TRUE, figures_dir = getwd())
```

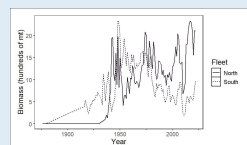
```
load(biomass_figure.rda)
```

```
rda$figure  
rda$caption
```

```
> "Biomass (B) time series. The horizontal dashed line  
represents the limit reference point (msy mt)."
```

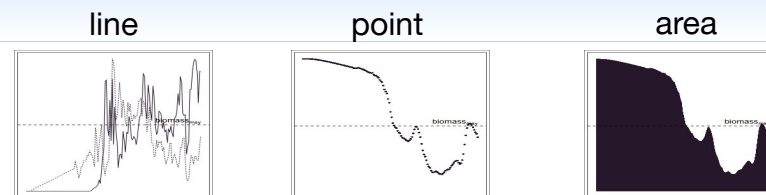
```
rda$alt_text
```

```
> "Line graph showing biomass time series. The x axis  
shows the year, which spans from 1874 to 2022. The y  
axis shows biomass in hundreds of mt, which spans from  
54.2 to 426.49."
```



Figures

time series



plot_biomass(dat, geom, ...)

Total biomass of line, point, or area over time with error where applicable.

plot_discard(dat, ...)

Observed vs. estimated discards by fleet over time.

plot_fishing_mortality(dat, ...)

Line or points of fishing mortality over time.

plot_landings(dat, ...)

Line, point, or area over time.

plot_natural_mortality(dat, ...)

Line graph of natural mortality over age classes.

plot_recruitment(dat, ...)

Observed and expected recruitment over time with error where applicable.

plot_selectivity(dat, ...)

Line graph of selectivity over age or length classes.

plot_spawning_biomass(dat, ...)

Line, point, or area over time with error where applicable.

plot_stock_recruitment(dat, ...)

Stock recruitment relationship with fitted line.

age series

plot_abundance_at_age(dat, z, proportional, ...)

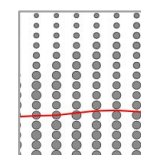
Bubble plot depicting abundance at age over time with trend lines across the x and y axes.

plot_biomass_at_age(dat, ...)

Bubble plot representing biomass at age over time.

plot_catch_comp(dat, ...)

Bubble plot representing catch at age over time with trend lines of top 5% strongest cohorts.



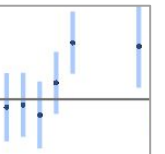
miscellaneous

plot_index(dat, ...)

Observed vs. estimated indices of abundance by fleet with error where applicable.

plot_recruitment_deviations(dat, ...)

Scatter plot of points with line segments depicting error.



Tables

Tables follow the same format and conventions of figures with the exception of facetting.

table_derived_quantities(dat, ...)

Specified derived quantities for management purposes. Default includes: SBmsy, Ftarget, R0, steepness.

table_indices(dat, ...)

Indices of abundance by fleet over time.

table_landings(dat, ...)

Total or indexed landings over time including error.

table_projections(dat, ...)

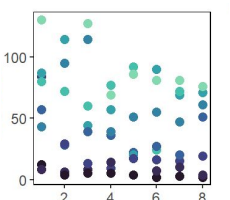
Projected spawning biomass, fishing mortality, and catch. Soon to be included in the {asar} report template Executive Summary.

Theming

theme_noaa()

Apply NOAA Fisheries theme to ggplot2 objects using (+) operator.

```
ggplot(OrchardSprays) +  
  geom_point(aes(x = rowpos, y = decrease,  
    color = treatment)) +  
  theme_noaa()
```



Exporting

save_all_plots(dat, ...)

Save each available plot and table in stockplotr as an rda in a folder called "figures".

Follow along with our planning and development on our [GitHub project](#) and [milestones](#)! Be on the lookout for future releases including more tables and figures!